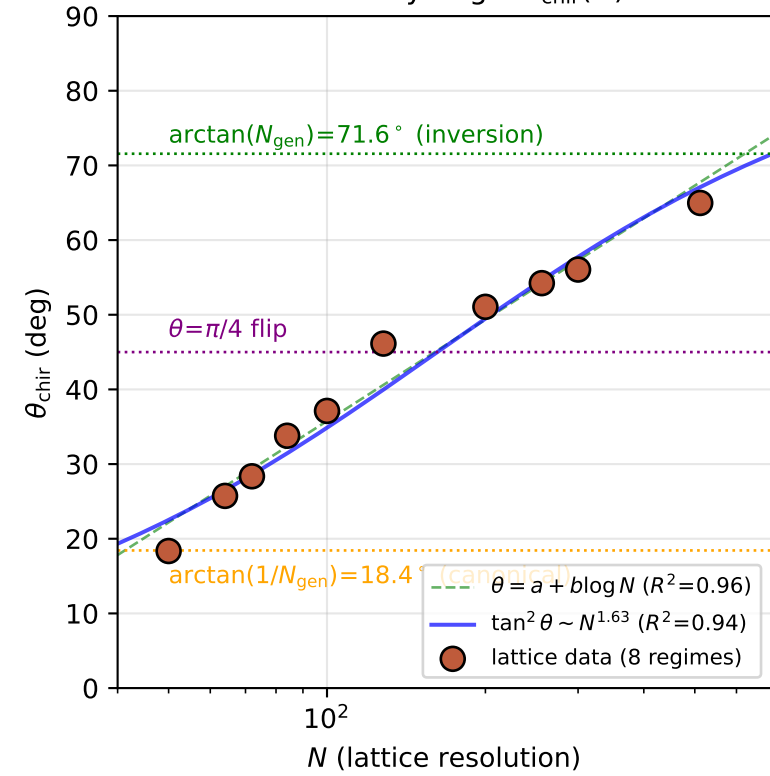
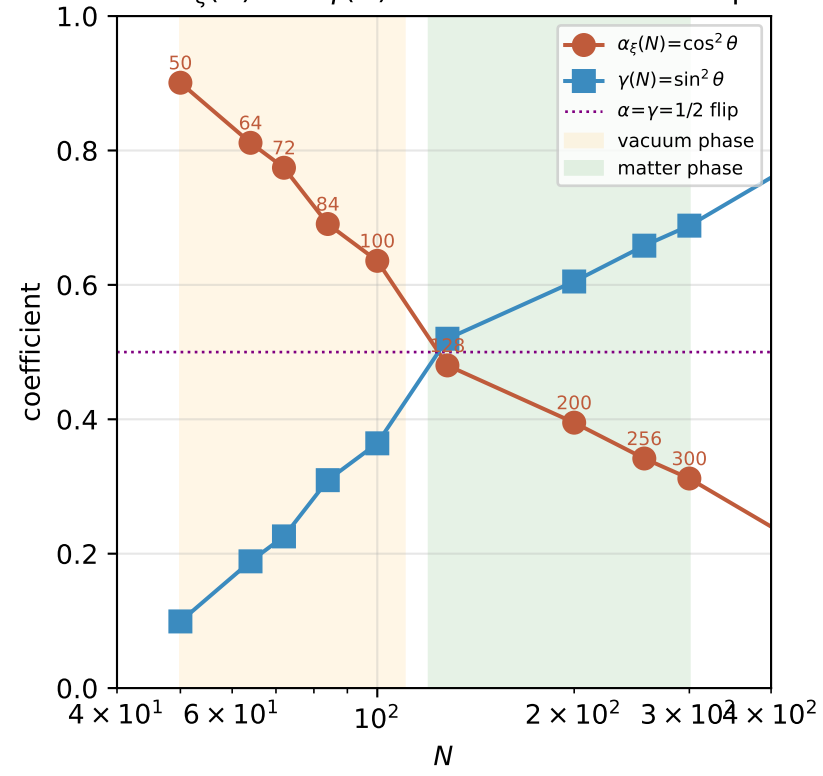


Chirality-flip phase diagram: dynamic $\theta_{\text{chir}}(N)$, vacuum-matter transition at $\theta=\pi/4$ (8-regime ladder $N \in [50, 300]$)

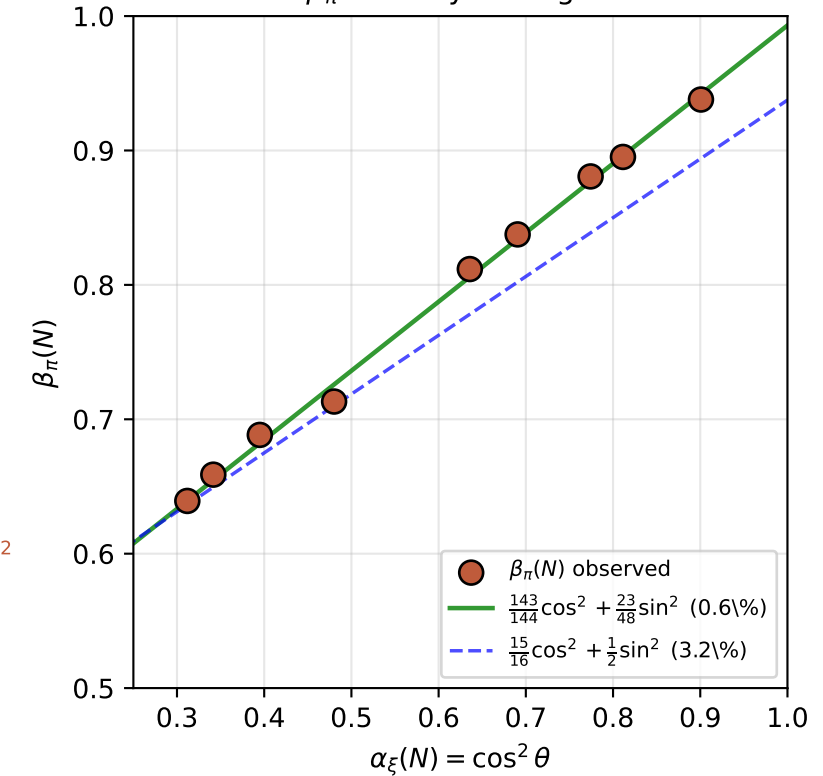
A. Chirality angle $\theta_{\text{chir}}(N)$



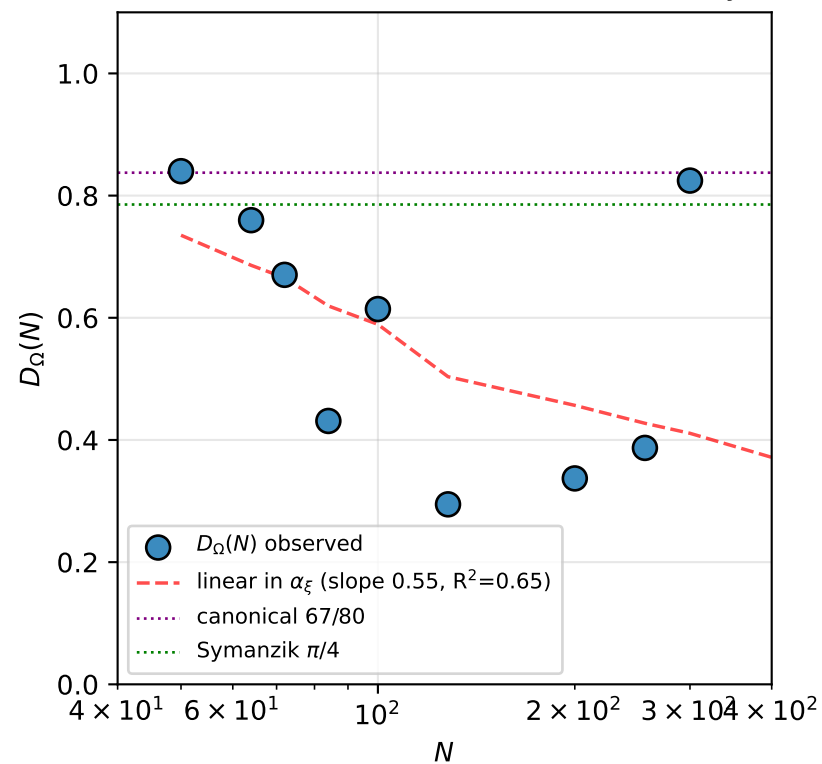
B. $\alpha_\xi(N)$ and $\gamma(N)$ with vacuum-matter flip



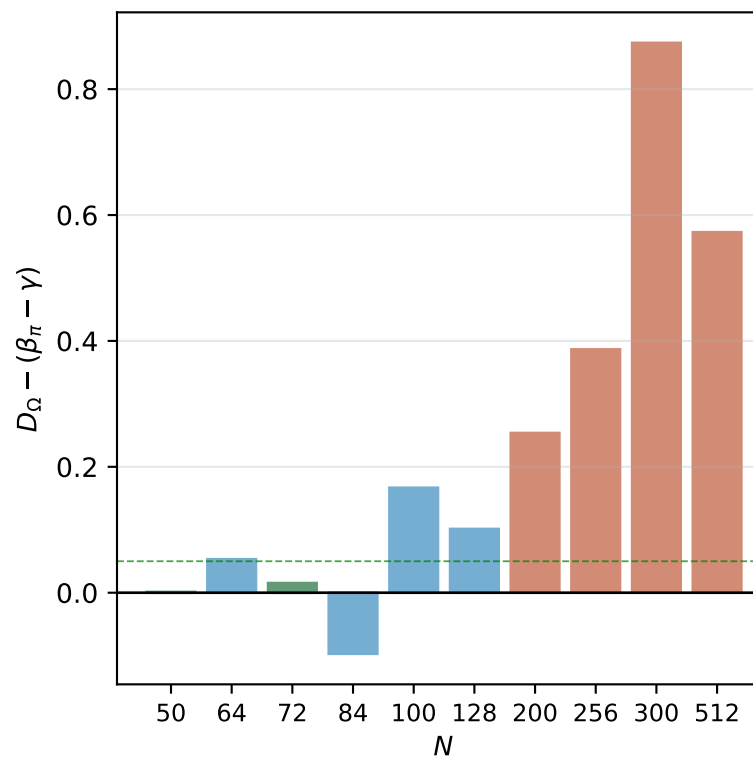
C. β_π chirality-mixing form



D. $D_\Omega(N)$: non-monotonic, NOT clean chirality-mix



E. C2 constraint residual: holds at vacuum, fails toward matter



F. $\alpha_\xi/\beta_\pi \sim N^{-2/5}$ clean power-law

